

An exact-arithmetic audit of hazard-rate ordering claims for finite mixtures

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Abstract

Hazard and reversed hazard rates of finite mixtures are ratios of two component sums. We audit a recurring argument that transfers a two-component ordering to n components because a T -transform leaves the other columns unchanged. The additive lifting lemma used in that argument is valid, but unchanged terms in a ratio do not cancel. Exact rational witnesses and certified sign changes refute several directly inspected n -component ordering statements. We separately test statements transmitted by later papers when the original text was unavailable; those results concern the transmitted formulation, not uninspected hypotheses in the original. Three-component exponential mixtures also answer negatively a proposed extension of a two-component hazard ordering. Further checks identify a distortion hypothesis that forces the family to be parameter-independent, and printed counterexample values inconsistent with exact evaluation. The evidence uses rational arithmetic and root counting, with outward-rounded interval certificates where stated. Each result is linked to its source status and reproduction program.

Keywords. finite mixture; hazard rate order; reversed hazard rate order; chain majorization; T -transform; Schur-convexity; exact computation; counterexample.

MSC 2020. 60E15; 62N05; 90B25.

1 Introduction

Finite mixtures with heterogeneous component parameters are the standard model of an inhomogeneous population in reliability and survival analysis, and comparing two such mixtures is usually organized by majorization: if the parameter vector of one mixture majorizes that of the other, the first population is “more heterogeneous”, and one asks which stochastic order this induces. For the usual stochastic order the answer is classical and additive — a mixture survival function $\sum_i p_i \bar{F}_i(t)$ is a sum over components, and a T -transform of the parameter matrix acts on each summand separately. For the hazard rate order the question is subtler: a mixture hazard rate

$$h_{p,v}(t) = \frac{\sum_{i=1}^n p_i v_i e^{-v_i t}}{\sum_{i=1}^n p_i e^{-v_i t}} \quad (1)$$

is a *ratio* of two sums, and a comparison that is true for two components does not, by any general principle, transport to n components.

Between 2018 and 2026 a family of papers asserted precisely such a transport. In various notations the claim is: if the $2 \times n$ matrix $\begin{bmatrix} p \\ \lambda \end{bmatrix}$ (weights over parameters) is related to $\begin{bmatrix} q \\ \gamma \end{bmatrix}$ by a single T -transform acting on two columns — or by a chain of such transforms — then the two n -component mixtures are ordered in the hazard rate or reversed hazard rate order, provided the two rows satisfy a comonotonicity or antiordering condition. The directly inspected proofs carrying this defect reduce the n -component comparison to the $n = 2$ case through the same step: the $n - 2$ untouched columns “stay fixed”, so only the affected pair matters. This note reports an exact-arithmetic audit of that literature. The step is invalid for a ratio of sums, and the conclusions it supports are false on certified instances.

*Prepared and reviewed with AI research agents. The accompanying scripts check algebraic identities and numerical certificates; see Section 10. These computational checks do not constitute a formal verification of the proofs.

What the audit found

- (a) **The defect and its reach** (Sections 2 and 6). The recurring step — we call it the *frozen-column lift* — is quoted verbatim in Section 2.5 from two open-access carriers; it appears in the proofs of hazard and reversed hazard rate theorems for ordinary mixtures, α -mixtures, distorted mixtures, location-scale mixtures and modified-proportional-hazard mixtures. The earliest source identified in the inspected citation record is Hazra and Finkelstein (2018) [19] and the standard citation for the additive criterion is the Lemmas 2.4/2.5 restatement in Barmalzan, Kosari and Balakrishnan (2022) [5], the citation handle used by most of the later papers.
- (b) **Certified counterexamples** (Section 4). Exact witnesses refute directly inspected n -component claims in [28] (both halves of Theorems 3.8 and 3.11, Theorems 3.9 and 3.12, Corollaries 3.2 and 3.3, and the $\alpha > 0$ branch of Theorem 3.4), [18] (Theorem 6, Corollaries 5 and 6), and [27] (Theorems 8, 9, 11, 12 and the dual Theorems 20–24, Corollaries 1–4). Further witnesses refute the transmitted formulations associated with [5] (Theorem 2), [19] (Theorem 3.4), and [25] (the extension of Theorem 4.2); attribution to the unavailable originals remains conditional. Sign certificates use rational evaluations, exact root counts, or outward-rounded intervals, as specified for each result. The PHR specializations associated with Theorem 11, Corollary 2 and Theorem 12 of [32] fail after a restrictive hypothesis is omitted; these are not counterexamples to the literal theorems.
- (c) **A resolved open question** (Section 5). Remark 6.18 of [33] (published Remark 6.3) asks whether the two-component hazard ordering extends to $n > 2$; [19] and [25] leave the same question open. The answer is no, already for ordinary mixtures of exponentials, by a certified three-component instance. A companion manuscript [17] analyses the failure for every $n \geq 3$.
- (d) **Degenerate or vacuous hypotheses** (Section 4). The hazard and reversed hazard theorems of [32] require a distortion-density monotonicity condition that no nontrivially γ -dependent family can satisfy ($\int_0^1 D'(u; \gamma) du = 1$). Parameter-independent families satisfy the condition, so the class is not empty, but the condition excludes nontrivial parameter variation. Two genuinely empty hypothesis classes occur in [13].
- (e) **Invalid printed counterexamples** (Section 4). Counterexamples printed as support for the theory are themselves wrong in two audited papers: [28] Counterexamples 3.7–3.8 and [10] Counterexamples 1–5 report values of order 10^{-13} – 10^{-16} at points where the exact difference is identically zero (floating-point noise reported as sign changes), or numbers contradicted by exact evaluation.
- (f) **Controls** (Section 6). The directly inspected two-component control theorems had no sampled violation, and every claim proved by the same lift on an *additive* functional — all audited stochastic-order theorems, where the mixture survival function is a sum — held on all admissible instances. The failure is specific to the rate functionals.

Scope and care. This is an audit, not a survey: we do not claim every paper in the citation cone is false (Section 7 lists the unaudited related-source list), we do not claim the underlying lemma is wrong (it is verified true for additive functionals), and we distinguish throughout between claims read verbatim in open texts and claims reconstructed from citing papers for paywalled sources. Sample counts reported as “holds-on-samples” are bounded-search statistics, never proofs.

2 The mathematics of the lift

2.1 Mixtures and orders

Let $p = (p_1, \dots, p_n)$ be a probability vector with positive entries. For a parametric family $\bar{F}(t \mid \lambda)$ of survival functions, the finite mixture with weights p and parameters $\lambda = (\lambda_1, \dots, \lambda_n)$ has survival function $S(t) = \sum_i p_i \bar{F}(t \mid \lambda_i)$. For nonnegative variables with survival functions S_X, S_Y , hazard rates h_X, h_Y and reversed hazard rates $r_X = f_X/F_X$, $r_Y = f_Y/F_Y$, write $X \leq_{\text{st}} Y$ if $S_X \leq S_Y$ everywhere, $X \leq_{\text{hr}} Y$ if $h_X \geq h_Y$ everywhere, and $X \leq_{\text{rh}} Y$ if $r_X \leq r_Y$ everywhere [31]; both rate orders imply the stochastic order. For the

exponential baseline $\bar{F}(t | v) = e^{-vt}$ the mixture hazard rate is (1), and with $y = e^{-t} \in (0, 1)$ its sign-relevant factor is the kernel

$$\tilde{h}_{p,\lambda}(y) = \frac{\sum_{i=1}^n p_i \lambda_i y^{\lambda_i}}{\sum_{i=1}^n p_i y^{\lambda_i}}, \quad (2)$$

a weighted ratio of two sums of power terms. The same kernel recurs across the cluster (Section 2.4).

2.2 Chain majorization and T -transforms

Write $[x; y]$ for the $2 \times n$ matrix with rows x and y . A T -transform is $T_\omega^{ij} = \omega I + (1 - \omega)\Pi^{ij}$, $\omega \in [0, 1]$, where Π^{ij} interchanges coordinates i and j ; right-multiplication replaces columns i, j by their ω -convex recombination and leaves the other $n - 2$ columns untouched. For vectors, $x \succ xT$ always, and $x \succ y$ iff $y = xT_1 \cdots T_k$ for finitely many T -transforms [23, Chapter 2.B]. The matrix analogue, (multivariate) *chain majorization* $A \gg B$ iff $B = AT_1 \cdots T_k$ acting on columns, is Chapter 15 of [23]. The hypotheses of the audited theorems restrict to the classes

$$\mathcal{V}_n = \left\{ \begin{bmatrix} x_1 & \cdots & x_n \\ y_1 & \cdots & y_n \end{bmatrix} : x_i, y_i > 0, (x_i - x_j)(y_i - y_j) \leq 0 \ \forall i, j \right\}$$

and $\mathcal{W}_n$, defined with ≥ 0 : the rows are antiorordered or comonotone. (Notation varies by paper: $\mathcal{A}_n/\mathcal{B}_n$ in [32], K_n/L_n in [18]; the content is identical.)

2.3 The additive lemma is true

Lemma 2.1 (the additive lift — valid). *Let $\Psi : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ be differentiable and let $\Psi_n(A) = \sum_{i=1}^n \Psi(a_{1i}, a_{2i})$ for a $2 \times n$ matrix A . Suppose the 2×2 criterion holds: $\Psi_2(A) \geq \Psi_2(B)$ for all $A \in \mathcal{V}_2$ (or $\mathcal{W}_2$) and $B = AT$. Then $\Psi_n(A) \geq \Psi_n(AT)$ whenever $A \in \mathcal{V}_n$ (or $\mathcal{W}_n$).*

This is Theorem 2 of Balakrishnan, Haidari and Masoumifard [3], restated as Lemmas 2.4 and 2.5 of [5] (the 2×2 criterion and the n -column lift respectively), Lemma 7 of [32] and Lemmas 1/2 and 3/4 of [18, 27]. It is correct: a T -transform on columns j, k leaves the remaining summands untouched, so the difference $\Psi_n(A) - \Psi_n(AT)$ reduces to a 2×2 difference, and the pairwise class membership needed by the criterion follows from $A \in \mathcal{V}_n$ or $\mathcal{W}_n$. Our audit verified the proof skeleton symbolically and confirmed the predicted directions on thousands of exactly generated admissible instances (Section 10). The lemma descends from the systems literature, where the compared functionals — series and parallel system survival or hazard — are additive or multiplicative over components [3, 15, 22].

2.4 The kernel is a ratio of sums

Every audited hazard or reversed hazard claim concerns one of the following functionals of the parameter matrix $[p; \lambda]$ (the exact model per paper varies; the algebraic shape does not):

Table 1: The ratio-of-sums kernel across the audited model families; in each line the ordering-relevant factor is a quotient of two additive column sums in a monotone parameter y or s .

Model family	rate functional (up to positive factors)
ordinary / PH mixtures	$h_{p,\lambda}(t) = \tilde{h}_{p,\lambda}(e^{-t})$, kernel (2)
α -mixtures [28, 33]	$h_U(t) = \theta^{-1} h(t/\theta) \tilde{h}_{p,\gamma}(\bar{G}(t/\theta)^\alpha)$, i.e. (2) at $y = \bar{G}^\alpha$
distorted mixtures [32]	$h_{P_n}(t) = g(t) \sum_i \pi_i D'(G; \gamma_i) / \sum_i \pi_i D(G; \gamma_i)$; (2) under $D(u; \gamma) = u^\gamma$
LS mixtures [5, 10, 35]	$h(s) = \sum_i p_i \nu_i s^{\nu_i} / \sum_i p_i s^{\nu_i}$ and $r(s) = \sum_i p_i \nu_i s^{\nu_i} / (1 - \sum_i p_i s^{\nu_i})$, $s = e^{-t}$
MPHR mixtures [18, 27]	$r_C = \lambda h(t) C(y)$ with $C(y) = [\sum_i p_i \alpha_i (1 - \bar{\alpha}_i y)^{-2}] / [\sum_i p_i \alpha_i (1 - \bar{\alpha}_i y)^{-1}]$, $y = \bar{F}^\lambda$

In every case the functional is a quotient $N(p, \lambda)/D(p, \lambda)$ of two additive column sums. Replacing two columns by their T -images leaves the summands of the other columns unchanged — but those summands sit inside both N and D , and the sign of the quotient's difference does not reduce to the two-column comparison. Formally:

Proposition 2.2 (the unchanged columns do not cancel). *Let (N_1, D_1) and (N_2, D_2) be the numerator and denominator contributions of the transformed pair, and let (a, b) be the contributions of the unchanged columns. For positive denominators,*

$$\frac{N_1 + a}{D_1 + b} - \frac{N_2 + a}{D_2 + b} = \frac{N_1 D_2 - N_2 D_1 + b(N_1 - N_2) + a(D_2 - D_1)}{(D_1 + b)(D_2 + b)}.$$

Thus the sign of the pair-only comparison does not in general determine the sign of the full comparison. The extra terms vanish only under additional conditions.

Proof. Expand the cross-multiplied numerator. The certificates below show sign reversals for admissible mixture parameters. $\square$

Consequently Lemma 2.1 is inapplicable. The necessary differential condition for the claimed T -monotonicity fails as well: at the certified $\mathcal{W}_n$ instance of Certificate 4.3, the n -column analogue of the 2×2 criterion evaluates to $25/11 > 0$ at $y = 1$ — the wrong sign.

2.5 The defect as printed

Two verbatim renderings of the step, from open-access texts:

“We observe that P_n and Q_n have the same distribution and same hazard rate function (i.e., $\pi_k = \pi_k^*$ and $\gamma_k = \gamma_k^*$) for all $k \neq i, j$ as the T-transform matrix $T_{\omega}^{i,j} = \omega I_n + (1 - \omega)\Pi^{i,j}$... Therefore, applying Theorem 8, the desired result follows immediately.” — proof of Theorem 11, [32]

“Let $\Phi_n(\theta) = h_{W_n}(t)$. It follows from Theorem 7 that Φ_2 satisfies Lemma 3. Hence, by applying Lemma 4, the desired result follows.” — proof of Theorem 8, [27]

Theorem 7/8 and Lemma 3/4 there are the $n = 2$ theorem and the additive lift; h_{W_n} is the n -component hazard rate, manifestly not of the form $\sum_i \Phi(a_{1i}, a_{2i})$. The same pattern occurs in [28], whose Theorem 3.8 proof sets $\Psi_n([p; \gamma]) = h_{U_n(p, \gamma, \theta)}(t)$ and invokes the additive Lemma 2.5, and in [18], where the n -component hazard theorem is asserted with *no* proof beyond “Similarly ... we can get the result”. The step is the frozen-column lift: the $n - 2$ unchanged columns are held fixed and the $n = 2$ theorem is applied to the remaining pair — invalid, because the frozen columns sit inside both the numerator and the denominator of the ratio (2).

In the finite grids searched, violations range from a few percent to over half of admissible instances (Section 4); these proportions describe those grids, not a distribution over all parameters. For each formulation refuted by a counterexample, repairing the proof requires changing a hypothesis or conclusion.

3 Audit method

Exact arithmetic. All parameters (weights, rates, scales, resilience parameters, transform weights ω , evaluation points) are rational; hypothesis predicates for each stated formulation — $\mathcal{V}_n/\mathcal{W}_n$ membership componentwise, antiordering classes $\mathcal{U}_n$, majorization and weak sub-/supermajorization, reciprocal majorization, T -transform factorization $[q; \gamma] = [p; \lambda]T$ — are verified exactly. Claimed ordering differences are reduced, after a monotone reparametrization ($s = e^{-t}$, $y = \bar{G}^\alpha$, $u = z^L$ clearing denominators), to rational functions whose sign changes are certified by exact rational evaluations of opposite sign together with exact real-root counts on the polynomial numerator: Sturm sequences via SymPy’s `Poly.count_roots` [7, 24] and isolation via the Vincent–Akritas–Strzeboński method (`Poly.intervals` [1]); where the degree exceeds the Sturm budget, outward-rounded interval arithmetic certifies the sign change instead. No uncertified floating-point comparison is used as a sign certificate; 60-digit evaluation is used only where genuinely algebraic quantities are compared, and is labelled *numerical*.

Classification vocabulary. Each claim receives one of: EXACT COUNTEREXAMPLE (certified sign change or exact opposite-sign witnesses), HOLDS-ON-SAMPLES (no violation on the exact-search domain; a bounded-search statistic, not a proof), VACUOUS (hypothesis class empty), DEGENERATE (hypotheses remove the parameter variation being compared), UNTESTABLE-BOUNDED (hypothesis class not exhausted) and NOT-CHECKABLE (commentary, or assumptions not reproducible from the text). For paywalled sources the scoreboard distinguishes *verbatim* statements, statements *transmitted* by citing papers, and the weakest consistent *claim shape*; a counterexample to a transmitted or claim-shape statement refutes that statement as cited, and the fidelity of the reconstruction is recorded per paper.

Admissible-instance generation. Searches draw rational instances from bounded grids (representative bounds: $n \in \{2, 3, 4\}$; weight numerators up to $1/40$ granularity; rates and scales among small rationals; $\omega \in \{\frac{1}{20}, \dots, \frac{19}{20}\}$; baseline families including exponential, Lomax, Pareto, power-law, proportional-reversed-hazard and compound Rayleigh), filtering to the hypotheses of the tested formulation before evaluating its conclusion. The SPBB PHR specialization explicitly omits the parameter-monotonicity hypothesis discussed below, so its counts are not counts of instances satisfying the literal theorem. Violation counts are reported with the admissible denominator.

Controls. Three control classes isolate the defect: (i) the $n = 2$ base theorems, which the lift generalizes (they are clean — e.g. 0/6133, 0/2264, 0/3200 violations in three independent two-component sweeps); (ii) the stochastic-order theorems proved by the same lift on additive functionals (clean — the lift is legitimate there); (iii) [33], audited for its main ordering claims, which posed the $n > 2$ hazard ordering as an open problem rather than claiming it.

4 Scoreboard and certificates

4.1 Scoreboard

Table 2 summarizes the audit; the “source” column records the strongest reading obtained (verbatim text / transmitted statement / claim shape only). Section 7 discusses the residual gap for non-verbatim rows.

4.2 Certificates

The six certificates below provide small exact witnesses and root counts for the stated formulations. Source reconstruction and omitted hypotheses are identified explicitly. Each is reproduced by a script named in Table 3.

Certified instance 4.1 (BKB2022 Theorem 2 claim shape; hazard order, $\mathcal{V}_3$). Ordinary exponential mixtures, integer rates $\nu = (13, 15, 11)$, weights $p = (\frac{3}{20}, \frac{1}{20}, \frac{4}{5})$, $[p; \nu] \in \mathcal{V}_3$; a single T -transform on columns $(1, 2)$ with $\omega = \frac{1}{2}$ gives $\nu' = (14, 14, 11)$, $p' = (\frac{1}{10}, \frac{1}{10}, \frac{4}{5})$. In $s = e^{-t}$ the hazard difference is rational with numerator of degree 29: exactly *two* roots in $(0, 1)$ (Sturm), with exact witnesses $\text{num}(\frac{5}{8}) < 0$ and $\text{num}(\frac{1}{16}) > 0$. The rates cross; neither direction of the claimed ordering holds. Sister certificates cover the rh claim, the class $\mathcal{W}_3$, the scale-row convention $[\lambda; p]$ with $\nu = 1/\lambda$ (interval-certified, denominator-positive numerators), and both directions simultaneously: the unconditional claim fails in every variant.

Certified instance 4.2 (HF2018 transmitted Theorem 3.4; ordinary mixtures, $n = 3$). $p = (\frac{1}{8}, \frac{29}{72}, \frac{17}{36})$, $\lambda = (9, 5, 4)$, so $[p; \lambda] \in \mathcal{V}_3$; a T -transform on columns $(1, 2)$ with $\omega = \frac{13}{20}$ gives $q = (\frac{2}{9}, \frac{11}{36}, \frac{17}{36})$, $\gamma = (\frac{38}{5}, \frac{32}{5}, 4)$, with $[q; \gamma] \in \mathcal{V}_3$ as well. With $w = u^{1/5}$ the kernel difference $\tilde{h}_{p, \lambda} - \tilde{h}_{q, \gamma}$ has a degree-83 numerator with exactly one root in $(0, 1)$ (Sturm) and exact witnesses of sign $+, -, -$ at $w = \frac{1}{2}, \frac{7}{8}, \frac{9}{10}$: the hazard rates cross once. The same instance refutes the reversed-hazard analog, and the corresponding two-component sweep (0/2264 crossings) confirms that the failure is specific to $n \geq 3$. This refutes the transmitted formulation when a one-factor chain is allowed; uninspected additional hypotheses in the original are not assessed.

Certified instance 4.3 (SKF2026 Theorem 3.8, $\mathcal{W}_n$ half; α -mixture). $p = (\frac{17}{44}, \frac{5}{11}, \frac{7}{44})$ and $\gamma = (7, 9, 2)$ comonotone, $[p; \gamma] \in \mathcal{W}_3$; T on columns $(1, 3)$, $\omega = \frac{17}{20}$, gives $q = (\frac{31}{88}, \frac{5}{11}, \frac{17}{88})$, $\delta = (\frac{25}{4}, 9, \frac{11}{4})$; exponential baseline, $\theta = 1$, $\alpha = -2$. On $y = \bar{G}^\alpha > 1$ the claimed $U_3 \leq_{\text{hr}} V_3$ reduces to $\tilde{h}_{p, \gamma} \leq \tilde{h}_{q, \delta}$; the difference has a numerator with exactly one root in the reduced variable and exact opposite-sign witnesses, e.g. the value

Table 2: Audit scoreboard. “Lift” marks the frozen-column T -transform defect; other defect mechanisms are named where different. Certified = exact rational/Sturm certificate exists.

Paper	Venue	Certified failures	Claim detail	Source
HF2018 [19]	TEST 27(4)	transmitted Thm 3.4 shape	lift; fails at $n = 3$ for ordinary mixtures; original proof unavailable	transmitted
BKB2022 [5]	PEIS 36(2)	Thm 2 shape	lift; certified crossings both directions, both classes, both conventions; $n = 2$ base claims fail at the unconditional shape	claim shape
NT2020 [25]	CSTM 51(10)	transmitted $n > 2$ extension of Thm 4.2	lift; the open question answered negatively	transmitted
GY2024 [18]	arXiv:2407.15638	Thm 6, Cors 5–6	lift; Thm 6 stated without proof; $\sim 48\%$ of admissible $n = 3$ instances violate	verbatim
BKF2024 [10]	MCAP 26(4)	Thms 2–6, 8; Lem. 1 misprinted; Cex 1–5, 7 invalid	other defects (wrong weak-majorization direction, active-subvector gap, missing hypothesis); the T -transform st theorems are clean	verbatim
VKF2025 [35]	Statistics 59(5)	st claims at $\alpha\gamma > 1$	different defect (a Schur-concavity step, not the lift); sharp boundary $\alpha\gamma = 1$: $0/\sim 10,200$ below, ~ 2050 above	claim shape
SKF2026 [28]	ASMBI 42(2)	Thms 3.4 ($\alpha > 0$), 3.8 both halves, 3.9, 3.10 (literal), 3.11 both halves, 3.12; Cors 3.2–3.3; Cex 3.7, 3.8 invalid	lift on the α -mixture hr/rh kernel	verbatim (AAM)
SPBB2026 [32]	JIA 2026:28	PHR specializations after dropping a restrictive hypothesis; literal family forced parameter-independent	lift, quoted verbatim; second defect: restrictive $-D'$ monotonicity	verbatim (OA)
SKB2026 [27]	Mathematics 14	Thms 8, 9, 11, 12, 20, 21, 23, 24; Cors 1–4	lift, quoted verbatim; dual MPRHR theorems share the kernels	verbatim (OA)
BTDK2024 [13]	arXiv:2412.10071	Thms 4.2, 4.3-E, 4.5-D; vacuity in 4.1, 4.4-E	not a carrier (no chain majorization); middle-region direction errors, a sign-indefinite product	verbatim
BGSK2025 [14]	arXiv:2511.00791	Thms 4.2, 4.3	not a carrier (vector majorization on the additive CDF only); a term-wise sign step fails	verbatim
SAF2022 [33]	PEIS 36(4)	none of the main claims; Thm 4.1(b) boundary failure; minor errata	clean control; posed the $n > 2$ question resolved in Section 5	verbatim (OA)

$-161379121858494821/28929682881644134876 < 0$ at $s = \frac{1}{4}$. Combined with the $\mathcal{V}_n$, $\alpha \geq 0$ half refuted in [17], both halves of Theorems 3.8 and 3.11 fail, and the chain claims Theorem 3.9, Theorem 3.12 and Corollaries 3.2–3.3 fail with them (a certified two-step $\mathcal{W}_3$ chain for Theorem 3.9 has numerator degree 855 and witnesses of both signs).

Certified instance 4.4 (SPBB2026 PHR specialization with the restrictive hypothesis omitted). $A = [\gamma; \pi] = (\frac{2}{7/12}, \frac{4}{1/12}, \frac{3}{1/3}) \in \mathcal{A}_3$ and $B = AT_{3/10}^{2,3} = (\frac{2}{7/12}, \frac{33/10}{31/120}, \frac{37/10}{19/120}) \in \mathcal{A}_3$, all memberships exact. Under $D(u; \gamma) = u^\gamma$ the hazard ratio kernel is $\tilde{h}_{\pi, \gamma}(u) = \sum_i \pi_i \gamma_i u^{\gamma_i} / \sum_i \pi_i u^{\gamma_i}$; in $z = u^{1/10}$ the difference $\tilde{h}_A - \tilde{h}_B$ has a degree-37 numerator with exactly two roots in $(0, 1)$ (Sturm) and witnesses $\tilde{h}_A - \tilde{h}_B = +108762382147/224621531888650 > 0$ at $z = \frac{1}{2}$ and a negative exact rational value at $z = \frac{15}{16}$. This instance violates the specialized conclusion associated with Corollary 2 (one-factor chain) and Theorem 12. It does not satisfy the literal parameter-monotonicity hypothesis and therefore is not a counterexample to those theorems as stated. In searches of the relaxed PHR formulation: 1725/5008 (resp. 2109/4106) admissible PHR instances violate the claimed direction at $n = 3$ (resp. $n = 4$), against 0/6133 at $n = 2$.

Certified instance 4.5 (GY2024 Theorem 6 (stated without proof); MPRHR mixture). $p = (\frac{16}{37}, \frac{9}{37}, \frac{12}{37})$, $\alpha = (\frac{1}{2}, \frac{8}{9}, \frac{2}{3})$ with $p_i \alpha_i = \frac{8}{37}$ for all i (the paper’s hypothesis, which makes the K_3 membership automatic); a single T -transform on columns $(1, 3)$ with $\omega = \frac{3}{4}$ gives $q = (\frac{15}{37}, \frac{9}{37}, \frac{13}{37})$, $\beta = (\frac{13}{24}, \frac{8}{9}, \frac{5}{8})$. The hazard-difference numerator $-6623y^7 + 217404y^6 - \dots$ has degree 7 and exactly two roots in $(0, 1)$ (Sturm), witnessed by $d(\frac{1}{16}) = -26382968577328/697805944384200125 < 0$ and $d(\frac{1}{5}) = 1056692120/7414054947243 > 0$: the claimed ordering

fails and the rates cross. Violation rates 1189/2500 ($n = 3$) and 766/1500 ($n = 4$); a certified two-step different-structure chain refutes Corollary 6.

Certified instance 4.6 (SKB2026 Theorem 8; MPHR α -mixture). $\alpha = -4$, $p = (\frac{7}{16}, \frac{1}{4}, \frac{5}{16})$, $\theta = (\frac{9}{4}, 9, 3)$, $[p; \theta] \in \mathcal{V}_3$; T on columns $(2, 3)$, $\omega = \frac{9}{10}$ gives $q = (\frac{7}{16}, \frac{41}{160}, \frac{49}{160})$, $\gamma = (\frac{9}{4}, \frac{42}{5}, \frac{18}{5})$. The hazard-difference numerator has degree 7 with two roots in $(0, 1)$ and witnesses $d(\frac{1}{100}) = -353619533609100/43666123426941601543 < 0$, $d(\frac{1}{4}) = 175665260436/20936322432487 > 0$. By the kernel identities of the model, the same instance refutes Theorems 11, 20, 21, 23, 24 and Corollaries 1–4 in matching pairs; Theorems 9 and 12 fall to certified two-step chains with intermediates in $\mathcal{V}_3$.

4.3 Beyond the lift: secondary certified findings

Three audited papers fail by other mechanisms, and two papers print invalid counterexamples:

- **Degenerate rate-theorem hypotheses in [32].** Every hr/rh claim requires $-D'(u; \gamma) = -\partial_u D(u; \gamma)$ monotone in γ at every $u \in (0, 1)$. Since $\int_0^1 D'(u; \gamma) du = D(1; \gamma) - D(0; \gamma) = 1$ for all γ , two such densities can be pointwise ordered only if they agree almost everywhere. The condition therefore forces $D(u; \gamma)$ to be independent of γ . It does not force $D = u$. Parameter-independent families remain admissible, so the hypothesis is degenerate rather than vacuous. Certificate 4.4 tests the advertised PHR specialization after this hypothesis is omitted. The paper’s own Example 1 hypothesis check is also false exactly (a claimed monotonicity fails on the example’s own parameter range).
- **The wrong-direction lemma in [10].** The printed Lemma 1 asserts $u \prec_w v \Rightarrow \varphi(u) \geq \varphi(v)$ for increasing Schur-concave φ ; the correct pairing is $\prec^w \leftrightarrow \geq$ for increasing Schur-concave (one-line refutation: $\varphi = \sum \sqrt{u_i}$). Theorems 2 and 8 invoke the false direction and are certified false as printed (Theorem 2 has 0/7366 sampled violations under the corrected $\prec^w$ reading — consistent with a typeset error). Theorems 3, 4, 5, 6 are certified false by missing hypotheses and by the piecewise-support/active-subvector gap: on bounded-support baselines the live components form a proper subvector while the Schur argument is applied to the full vector. Theorem 10’s conclusion holds on all 85,410 admissible evaluations but its key intermediate step (equation (49)) is certified false — an unproven theorem whose statement may still be true. Its T -transform stochastic-order theorems (11–13) apply the additive lift correctly and are clean on ~ 7000 evaluations.
- **Inconsistent printed counterexample numerics in [10]** (Counterexamples 1, 2, 3, 5 report $\sim 10^{-13}$ – 10^{-16} “sign changes” at parameter points where only one component is alive and the difference is identically zero; Counterexample 4’s printed -3.45×10^{-6} is contradicted by the exact $+5.19 \times 10^{-7}$; Counterexample 7’s claimed sign change is absent on $\sim 12,000$ exact rational points) and in [28] (Counterexample 3.8’s $\pm 10^{-16}$ values at points where $h_U \equiv h_V = \gamma/(2t)$ identically; Counterexample 3.7’s printed -0.0834 where the exact value is $+0.0674$).
- **Vacuity and direction defects outside the lift family in [13]** (the IPRFR/IRFR proportional-ageing hypotheses are unsatisfiable on unbounded support; certified failures of Theorems 4.2, 4.3-E, 4.5-D — including the cluster’s first broken *stochastic-order* claim) and [14] (certified failures of the two multiple-outlier rate theorems via a term-wise sign step invalid when $f'/f < 0$; plus an independent definition-vs-theorem direction inconsistency).

5 The resolved open question

Both [19] (§4) and [25] (§4) leave the $n > 2$ hazard-rate ordering open, as does Remark 6.18 (published Remark 6.3) of [33]: whether the two-component majorization comparison of mixture hazard rates extends to $n > 2$ components. The weakest form of the claim — ordinary exponential mixtures, common weights, vector majorization — already fails:

Certified instance 5.1 ($n > 2$ open question, negative). $p = (\frac{39}{86}, \frac{18}{43}, \frac{11}{86})$ (decreasing), $\lambda = (4, 6, 9)$ and $\gamma = (4, 7, 8)$ (increasing), so (p, λ) and (p, γ) lie in $\mathcal{U}_3$ and $\lambda \succ \gamma$ (partial sums $4 \leq 4$, $10 \leq 11$, $19 = 19$), all

verified exactly. With $s = e^{-t}$ the difference $\tilde{h}_{p,\lambda} - \tilde{h}_{p,\gamma}$ has a degree-9 numerator, a denominator nonvanishing on $(0, 1)$, exactly one interior root (Sturm), and exact witnesses

$$d(\tfrac{1}{2}) = \frac{6271}{156247} > 0, \quad d(\tfrac{3}{4}) = -\frac{17228781}{103906915} < 0, \quad d(1) = -\frac{25}{86} < 0.$$

The hazard rates cross once: the mixtures are hr-incomparable, and no consistent $n > 2$ ordering exists under these hypotheses. The companion manuscript [17] proves the failure for every $n \geq 3$ with equal weights and integer rates, identifies the mechanism (the tail is governed by the second-smallest rate, which majorization does not control in the needed direction), and shows what survives: the stochastic order, the ordering near the origin, and the two-component statement.

6 Provenance

The citation cone. The lift is genuinely Schur-theoretic: Marshall–Olkin–Arnold chain majorization reduces a $2 \times n$ comparison to T -transforms, and for additive functionals the reduction to $n = 2$ is the standard argument, developed for series/parallel systems by [15, 22] and stated as the canonical $2 \times n$ criterion in Theorem 2 of [3] — valid there, because system hazard and survival functionals are sums or products over components. Hazra, Kuiti, Finkelstein and Nanda [20] carried the criterion to location-scale max order statistics, still additive. The defect is born at the systems→mixtures boundary.

Earliest identified attribution: 2018. Hazra and Finkelstein [19] is the earliest paper identified in this audit as importing multivariate chain majorization into mixture-model hr/rh comparisons. The full text is paywalled and was not retrieved (Section 7), but two independent open-access witnesses pin the n -component T -chain hazard claims to its Section 3: the accepted manuscript of [28] states verbatim “The proof follows from Theorem 3.4 of Hazra and Finkelstein (2018)” for both of its different-structures chain theorems (3.9 and 3.12), and the theorem-number map in [32] (Remarks 8, 11, 12 and Table 1) places the n -component single- T and T -chain hr claims at HF2018 Theorems 3.6/3.8/3.10/3.12; [27] likewise routes its different-structure theorems to “Theorem 3.4 of [Hazra and Finkelstein (2018)]”. The transmitted statement is refuted by Certificate 4.2. This does not identify the argument used in the unavailable original proof or rule out omitted hypotheses; the proof-level attribution remains unresolved.

Later citations. [5] (online September 2020) restated the additive criterion as Lemmas 2.4/2.5, which later papers cite. Certificate 4.1 refutes the reconstructed unconditional rate-ordering formulation; the original hypotheses remain to be checked. Related papers appeared in 2020–2021, including [4, 25, 26, 30]. The directly inspected 2024–2026 papers supply the quoted uses of the additive lift. Citation links establish a route for further checking; they do not establish that every citing paper uses the invalid step.

Clean controls. The audit’s controls confirm the diagnosis rather than a blanket failure: the directly inspected $n = 2$ control theorems held on all sampled admissible instances; every claim proved by the lift on an additive functional — the stochastic-order theorems of [10] (Theorems 11–13), [18] (Theorems 2 and 4), [27] (Theorems 1–6) and the st claims of [5] at mechanism level — is legitimate and consistent; and [33], which posed the $n > 2$ question rather than answering it, audited clean apart from a boundary failure of one biconditional (its Theorem 4.1(b), repaired by a strict inequality) and minor errata. The systems literature above the cone is unaffected.

7 Limitations

- **Non-verbatim rows.** [5, 19, 25, 35] are paywalled; their rows in Table 2 are audited through transmitted statements or reconstructed formulation, whose fidelity to every original hypothesis has not been established. The certificates stand against the claim shape in all cases; the attribution of an exact theorem number is as recorded. For [19] the verbatim proof text remains unretrieved, and only the later attribution, not the original proof, has been checked.

- **Holds-on-samples is not a proof.** Consistent results are bounded-search statistics over the parameter families listed in each memo; they are reported with admissible counts and are never conflated with proofs. The certified failures, conversely, are proof-grade.
- **Related sources are not findings of error.** Roughly a dozen further papers share the signature (n -component hr/rh claims under chain or matrix majorization): [4, 8, 9, 11, 12, 21, 26, 29, 30, 34] among them; papers are named from claim shape alone until audited, and membership of the queue is not a finding of falsity. Two related sources subsequently audited, [13, 14], turned out not to carry the lift at all (they fail, or pass, by other mechanisms).
- **Violation rates are sampling artifacts.** The percentages quoted (e.g. $\sim 48\%$ for [18] at $n = 3$) describe the exact-search distribution, not a property of the claims; the certified instances are the content.
- **Degree budget.** A few certificates use outward-rounded interval arithmetic in place of a Sturm count where the numerator degree exceeded the certification budget (e.g. the degree-3325 numerator of the location-scale specialization certificate); the sign change is certified by enclosure bounds; this is distinct from an exact Sturm root count.

8 What correct statements look like

The audit localizes the failure, which suggests the shape of valid replacements:

1. **Additive targets.** The lift is theorem-grade for functionals additive over columns: mixture survival and distribution functions under the corresponding additive hypotheses, and, on the systems side, sums and products of component rates [3]. Any n -component rate-ordering claim must either be re-argued on an additive surrogate or carry hypotheses strong enough to control the ratio (2) directly.
2. **What a true n -component hazard ordering would need.** The mechanism analysis of [17] shows the obstruction is structural: majorization controls the $t \rightarrow 0$ behavior (equal means, larger variance) but the tail is governed by the minimal and second-minimal rates, which majorization does not order in the required direction. A correct theorem would need hypotheses fixing the tail — e.g. equalizing the smallest two rates jointly, or conditions on the whole vector rather than on pairs of columns — and even equalizing weighted means does not suffice (Certificate 5.1 has unequal p -means at $t = 0$, but certified sign changes also occur away from the origin). No repair by a distributional-class restriction exists for the lift claims (the failures use the exponential baseline, the common specialization of every model).
3. **The honest boundary results.** For the equal-weight exponential-mixture model studied in the companion paper, the usual stochastic order and the local hazard comparison near $t = 0$ survive under its stated hypotheses. Its two-component hazard result also survives. These conclusions are not blanket claims for every model in the audit. Authors repairing the affected theorems should expect the correct n -component statements to be weaker than the printed ones.
4. **Reading list for reviewers.** The $\int D' = 1$ constraint that removes parameter variation in [32], the unsatisfiable proportional-ageing hypotheses of [13], and the misprinted weak-majorization lemma of [10] are checkable by hand. The invalid printed counterexamples show why exact evaluation is useful when reported differences are near rounding error.

9 Related work and conclusion

The mathematics sits inside stochastic-order theory [23, 31]. Mixture-ordering work by other techniques — component-wise ordering rather than parameter-matrix majorization — is outside the defect cone (e.g. [2, 6, 16] and the Navarro line). The certification methodology — exact rational witnesses plus Sturm/Vincent-Akritas-Strzeboński root counts [1, 7, 24] — is the same discipline used in [17], applied there to a single theorem and here to a literature cluster.

The central obstruction is algebraic: unchanged component contributions remain in the numerator and denominator of a mixture rate. A valid additive lemma therefore cannot supply the claimed reduction on its

own. The directly checked counterexamples demonstrate failures of specific printed conclusions, while the re-constructed rows identify precise statements for comparison with their unavailable originals. Rational witnesses and explicit source status make these findings independently testable.

10 Data availability

The accompanying repository (<https://github.com/hraness/algal-lab>) contains the verification programs mapped in Table 3. Most are under `research/spikes/`; the formerly local BKF and additive-lemma programs are now included under `papers/mixture-cluster-audit/verify/`. Certificate assertions and printed exact values should be checked together; exploratory sweeps are supporting evidence, not proof checkers. All use SymPy 1.14 exact arithmetic (rational `Rational/Fraction` values, `Poly.count_roots` and `Poly.intervals`) with `mpmath` interval arithmetic where noted. Table 3 maps the claims of this paper to their reproductions; per-paper audit memos (`memo.md` in each directory) record the full per-claim status, hypothesis predicates, search bounds and retrieval logs.

Table 3: Reproduction map. Scripts run with any Python 3.12 + SymPy 1.14 interpreter; e.g. `python3 certify_thm11.py`. Paths are relative to the `research-spikes` directory; the prefix `paper:` denotes this paper’s verification supplement.

Claim or check	Script (<code>research/spikes/...</code>)	Key exact witnesses
SKF Thm 3.4 ($\alpha > 0$)	<code>mixture-audit/certify_thm34.py</code>	$d(1) = \frac{305}{3888} > 0$; $d(20) = \frac{-1579547560}{2598836427243} < 0$
SKF Thm 3.8/3.11 $\mathcal{W}$ -halves; Thm 3.9 $\mathcal{W}$ (2-step)	<code>mixture-audit/certify_whalf.py</code>	$\frac{-161379121858494821}{28929682881644134876} < 0$ at $s = \frac{1}{4}$; deg-855 numerator, alternating-sign witnesses
SKF/SAF full claim tables	<code>mixture-audit/: audit_skf_st, audit_skf_hr, audit_saf, verify_paper_cex</code>	admissible counts per memo; Cex 3.7/3.8 refutations
SPBB PHR test with hypothesis omitted (Cert. 4.4)	<code>shekari-audit/certify_thm11.py</code>	+108762382147/ 224621531888650; Sturm: 2 roots in $(0, 1)$
SPBB degeneracy; Example 1 check SPBB LS specialization (= BKB Thm 2 map)	<code>shekari-audit/audit_shekari.py</code> <code>shekari-audit/certify_ls.py</code>	$\partial_\gamma(-D')$ sign changes, exact opposite-sign interval witnesses at $z = 999/1000, 999/2000$
BKB Thm 2 shape, 4 cells (Cert. 4.1)	<code>bkb2022-own/: certify_int, certify_iv, certify_n2</code>	Sturm root counts 3, 3, 2, 2 in $(0, 1)$; scale-row interval certs
HF transmitted Thm 3.4 (Cert. 4.2)	<code>hf2018-nt2020/certify_hf_T_n3.py</code>	deg-83 numerator, 1 root in $(0, 1)$; signs +, −, −
Open question (Cert. 5.1)	<code>hf2018-nt2020/certify_nt42_n3.py</code>	$6271/156247 > 0$; $-17228781/103906915 < 0$; 1 interior root
VKF2025 R1 ($\alpha\gamma > 1$)	<code>vkf2025/certify_r1.py</code>	$d(12) = \frac{19760252621}{571952479518784} > 0$; $d(15) < 0$; 1 root in $(12, 13)$
GY Thm 6/Cor 5–6; SKB Thms 8–24, Cors 1–4 (Certs. 4.5, 4.6)	<code>gy-skb-audit/: certify, certify_chains</code>	deg-7/9/13 numerators, 2 roots each; witnesses per memo
BKF Thms 2–8; Lemma 1; Cex 1–5, 7; Thm 10 step	<code>paper:bkf2024/: certify, scan_all, sturm_cert, verify_cex, verify_rest</code>	whole-cell cert: $h_U - h_V > 0$ on all of $(\frac{7}{20}, \frac{43}{30}]$
BTDK Thms 4.2, 4.3-E, 4.5-D	<code>btdk-arxiv/audit_btdk.py</code>	exact middle-region witnesses; algebraic lr cert
BGSK Thms 4.2, 4.3	<code>bgs-k-arxiv/audit_bgsk.py</code>	$\zeta(4) = 455/344 > \zeta(5) = 535/406$; $\Lambda(20) < \Lambda(80)$
Lemma validity + non-separability (Prop. 2.2)	<code>paper:bhb2022/: audit_bhb, sweep_bhb</code>	$\tilde{h}(A) - \tilde{h}(AT)$ changes with untouched column; $M_{13}(1) = 25/11 > 0$
Additive-lift controls (st theorems)	<code>gy-skb-audit/check_st.py</code> ; <code>mixture-audit/audit_skf_st.py</code>	0 violations on all admissible st claims

The audit summary (Table 2), the citation map and the retrieval logs are in the `cluster-audit` directory of the same tree. Recorded execution environment: SymPy 1.14.0, mpmath 1.3.0, Python 3.12.

References

- [1] Alkiviadis G. Akritas and Adam W. Strzeboński. A comparative study of two real root isolation methods. *Nonlinear Analysis: Modelling and Control*, 10(4):297–304, 2005. doi: 10.15388/NA.2005.10.4.15110.
- [2] M. Amini-Seresht and Y. Zhang. Stochastic comparisons on two finite mixture models. *Operations Research Letters*, 45(5):475–480, 2017.
- [3] Narayanaswamy Balakrishnan, Abedin Haidari, and Kambiz Masoumifard. Stochastic comparisons of series and parallel systems with generalized exponential components. *IEEE Transactions on Reliability*, 64(1):333–348, 2015. doi: 10.1109/TR.2014.2354192.
- [4] Ghobad Barmalzan, Sajad Kosari, and Zhengcheng Zhang. Ordering results for finite α -mixtures. *Statistics & Probability Letters*, 173:109083, 2021. doi: 10.1016/j.spl.2021.109083.
- [5] Ghobad Barmalzan, Sajad Kosari, and Narayanaswamy Balakrishnan. Orderings of finite mixture models with location-scale distributed components. *Probability in the Engineering and Informational Sciences*, 36(2):461–481, 2022. doi: 10.1017/S0269964820000467.
- [6] Jarosław Bartoszewicz and Magdalena Skolimowska. Preservation of stochastic orders under mixtures of exponential distributions. *Probability in the Engineering and Informational Sciences*, 20(4):655–666, 2006. doi: 10.1017/S0269964806060402.
- [7] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, Berlin, Heidelberg, second edition, 2006. doi: 10.1007/3-540-33099-2.
- [8] Raju Bhakta, Narayanaswamy Balakrishnan, Suchandan Kayal, and B. Pradhan. Stochastic comparisons of finite mixtures with generalized weibull components. *Statistics*, 58(3):552, 2024.
- [9] Raju Bhakta, Suchandan Kayal, and Narayanaswamy Balakrishnan. Ordering results between two multiple-outlier finite δ -mixtures. *Statistics & Probability Letters*, 213:110193, 2024. doi: 10.1016/j.spl.2024.110193.
- [10] Raju Bhakta, Suchandan Kayal, and Maxim Finkelstein. Stochastic comparisons for finite mixtures from location-scale family of distributions. *Methodology and Computing in Applied Probability*, 26(4):52, 2024. doi: 10.1007/s11009-024-10121-x.
- [11] Raju Bhakta, A. Kundu, Suchandan Kayal, and M. Alizadeh. Ordering results between two finite mixture models with inverse kumaraswamy components. *Mathematics*, 12(6):852, 2024. Also arXiv:2311.17568.
- [12] Raju Bhakta, Priyanka Majumder, Suchandan Kayal, and Narayanaswamy Balakrishnan. Ordering results between two finite mixture models with general distributed components. *Metrika*, 87(6):681–712, 2024.
- [13] Raju Bhakta, Nuria Torrado, Subrata Das, and Suchandan Kayal. Ordering results between two finite arithmetic mixture models with multiple-outlier location-scale distributed components, 2024. URL <https://arxiv.org/abs/2412.10071>. arXiv:2412.10071v1 [math.ST], 13 December 2024.
- [14] Raju Bhakta, K. Gupta, G. Saadat Kia (Barmalzan), and Suchandan Kayal. Stochastic comparisons of finite mixtures with general exponentiated location-scale distributed components, 2025. URL <https://arxiv.org/abs/2511.00791>. arXiv:2511.00791v1 [math.ST], 2 November 2025.
- [15] Richard Dykstra, Subhash Kochar, and Javier Rojo. Stochastic comparisons of parallel systems of heterogeneous exponential components. *Journal of Statistical Planning and Inference*, 65(2):203–211, 1997. doi: 10.1016/S0378-3758(97)00058-X.
- [16] Maxim Finkelstein and Veronica Esaulova. Asymptotic behavior of a general class of mixture failure rates. *Communications in Statistics - Theory and Methods*, 35(11), 2006.
- [17] Benjamin Guo. Majorization does not order the hazard rates of exponential mixtures with more than two components, 2026. Companion manuscript.
- [18] Lina Guo and Rongfang Yan. Orderings of the finite mixture with modified proportional hazard rate model, 2024. URL <https://arxiv.org/abs/2407.15638>. arXiv:2407.15638v2 [math.ST], 23 July 2024.
- [19] Nil Kamal Hazra and Maxim Finkelstein. On stochastic comparisons of finite mixtures for some semiparametric families of distributions. *TEST*, 27(4):988–1006, 2018. doi: 10.1007/s11749-018-0581-7.
- [20] Nil Kamal Hazra, Mithu Rani Kuiti, Maxim Finkelstein, and Asok K. Nanda. On stochastic comparisons of maximum order statistics from the location-scale family of distributions. *Journal of Multivariate Analysis*, 160:31–41, 2017.
- [21] Suchandan Kayal, Raju Bhakta, and Narayanaswamy Balakrishnan. Ordering results between two finite mixture models. *Stochastic Models*, 39(2):363–382, 2023.
- [22] Baha-Eldin Khaledi and Subhash Kochar. Some new results on stochastic comparisons of parallel systems. *Journal of Applied Probability*, 37(4):1123–1128, 2000. doi: 10.1239/jap/1014843095.
- [23] Albert W. Marshall, Ingram Olkin, and Barry C. Arnold. *Inequalities: Theory of Majorization and Its Applications*. Springer Series in Statistics. Springer, New York, second edition, 2011. doi: 10.1007/978-0-387-68276-1.

- [24] Aaron Meurer, Christopher P. Smith, Mateusz Paprocki, Ondřej Čertík, Sergey B. Kirpichev, Matthew Rocklin, AMiT Kumar, Sergiu Ivanov, Jason K. Moore, Sartaj Singh, Thilina Rathnayake, Sean Vig, Brian E. Granger, Richard P. Muller, Francesco Bonazzi, Harsh Gupta, Shivam Vats, Fredrik Johansson, Fabian Pedregosa, Matthew J. Curry, Andy R. Terrel, Štěpán Roučka, Ashutosh Saboo, Isuru Fernando, Sumith Kulal, Robert Cimrman, and Anthony Scopatz. SymPy: symbolic computing in Python. *PeerJ Computer Science*, 3:e103, 2017. doi: 10.7717/peerj-cs.103.
- [25] Hossein Nadeb and Hamzeh Torabi. New results on stochastic comparisons of finite mixtures for some families of distributions. *Communications in Statistics - Theory and Methods*, 51(10):3104–3119, 2022. doi: 10.1080/03610926.2020.1788082. Published online 9 July 2020.
- [26] K. Panja, A. Kundu, and B. Pradhan. Stochastic comparisons of finite mixture models with proportional odds and related families. *Stochastic Models*, 38(2):190–213, 2022.
- [27] Supriya Sahoo, Suchandan Kayal, and Narayanaswamy Balakrishnan. Ordering results between two finite α -mixture models with components following modified proportional hazard rate model. *Mathematics*, 14(14):2557, 2026. doi: 10.3390/math14142557.
- [28] Supriya Sahoo, Suchandan Kayal, and Maxim Finkelstein. Stochastic ordering results between two finite α -mixture models with resilience-scaled components. *Applied Stochastic Models in Business and Industry*, 42(2):e70089, 2026. doi: 10.1002/asmb.70089. Statement numbers and wording cited here refer to the peer-reviewed accepted author manuscript deposited in the University of Strathclyde repository (Strathprints record 96227).
- [29] Supriya Sahoo, Neeraj Misra, and Nil Kamal Hazra. Stochastic comparisons of α -mixtures of survival functions and related models. *Statistics*, 2026. doi: 10.1080/02331888.2026.2661806.
- [30] M. Sattari, Ghobad Barmalzan, and Narayanaswamy Balakrishnan. Stochastic comparisons of finite mixtures with general semiparametric components. *Communications in Statistics - Theory and Methods*, 2021. doi: 10.1080/03610926.2021.1880592.
- [31] Moshe Shaked and J. George Shanthikumar. *Stochastic Orders*. Springer Series in Statistics. Springer, New York, 2007. doi: 10.1007/978-0-387-34675-5.
- [32] M. Shekari, Z. Pakdaman, G. Saadat Kia Barmalzan, and N. Balakrishnan. Stochastic comparisons of finite mixture models derived from distorted distributions: results based on vector and multivariate chain majorization approaches. *Journal of Inequalities and Applications*, 2026:28, 2026. doi: 10.1186/s13660-026-03450-7.
- [33] Omid Shojaee, Majid Asadi, and Maxim Finkelstein. Stochastic properties of generalized finite α -mixtures. *Probability in the Engineering and Informational Sciences*, 36(4):1055–1079, 2022. doi: 10.1017/S0269964821000243. Published online 15 July 2021.
- [34] Omid Shojaee, Majid Asadi, and Maxim Finkelstein. Hazard rate shape and related ageing properties of generalized finite α -mixtures. *Communications in Statistics - Theory and Methods*, 53(11):4062–4084, 2023.
- [35] Akhila Anna Varghese, Ameen Mahmood K., Soumyadeep Sarkar, Shyamal Ghosh, and Priyanka Majumder. On stochastic comparisons of α -mixture models with location-scale family of distributions. *Statistics*, 59(5):1278–1300, 2025. doi: 10.1080/02331888.2025.2502946.